Statistics for Computer Scientists

Lecture 11: Multiple regression

28 November 2025

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Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Nonparametric statistics	Chapter 15
12 Dec	Assignment discussion	

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Today's lecture

- Recap lecture 10 (regression and correlation)

- Multiple explanatory variables
- Multiple correlation and R^2
- Infererence with multiple regression
- Residual plots
- Interaction and categorical predictors
- Logistic regression

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Recap regression and correlation

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Regression and correlation

- Population regression function $\mu_y = \alpha + \beta x$ describes mean of Y at specific value of X, conditional standard deviation describes variability of Y given X
- Prediction equation $\hat{y} = a + bx$
- Correlation is standardized slope: $r = b\left(\frac{s_X}{s_Y}\right)$
- Residual / regression / total sum of squares
- r-squared

Multiple explanatory variables

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Bivariate -> multivariate

- Bivariate analyses are informative, but usually a first step
- Many explanatory variables may have an association with the response variable
- The effect of an explanatory variable on a response variable may change when we take other explanatory variables into account!

- *Example: consider X = height and Y = score on math test (observations are children). Using a linear model, you will find an association between X and Y. But if we also take grade level into account, the association will disappear!*

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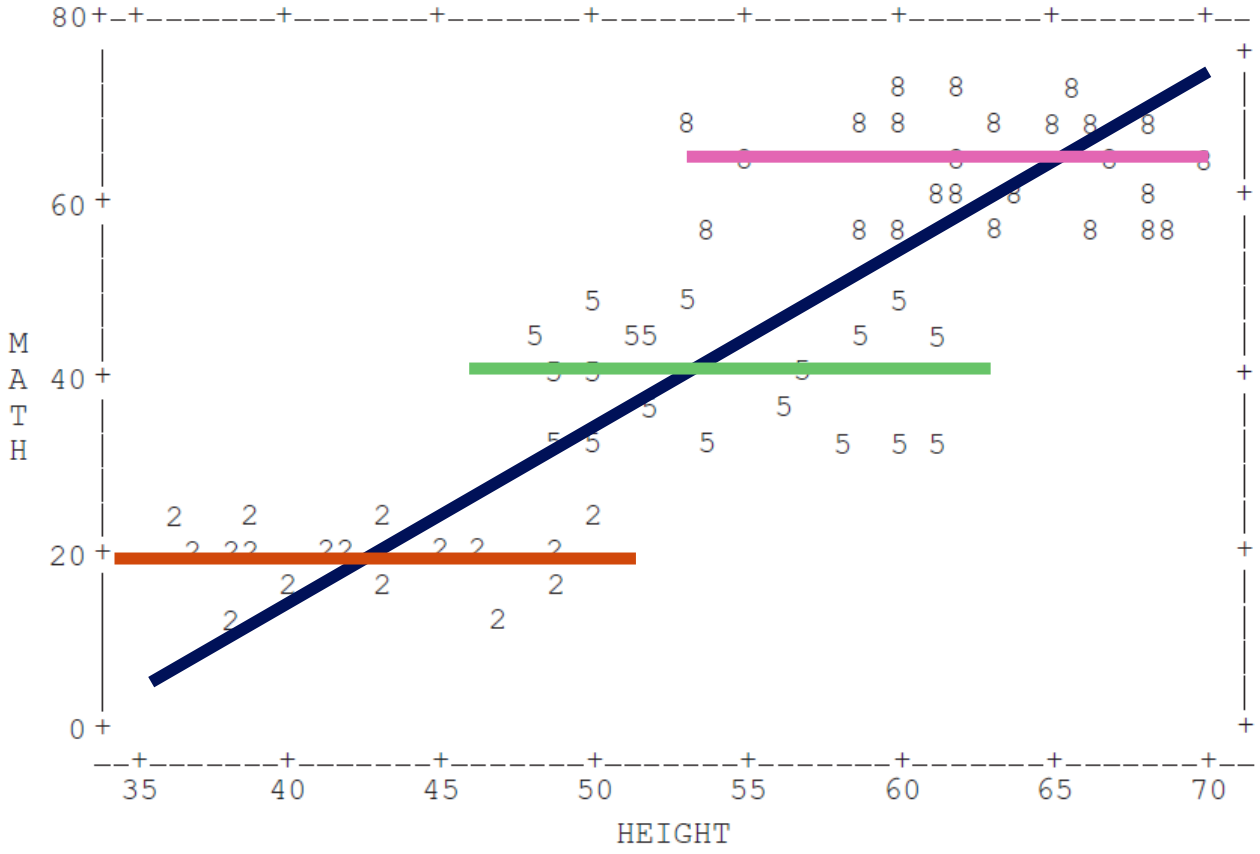
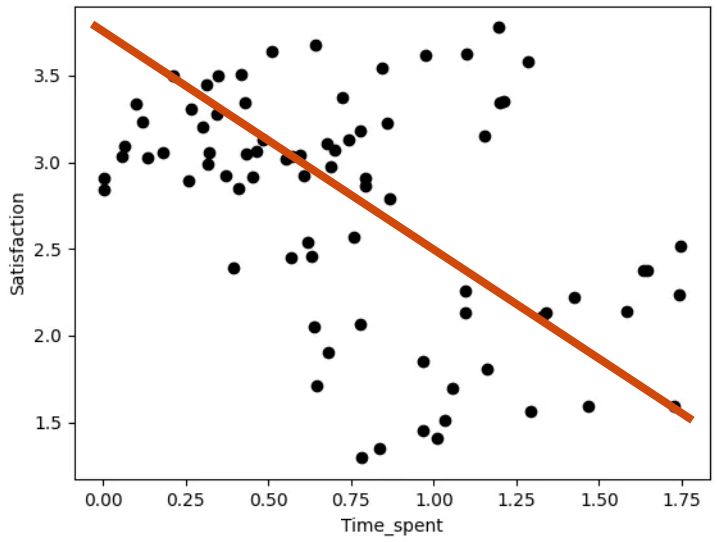
Example

FIGURE 10.1: Scatterplot Showing Relationship between Height and Math Achievement Test Score, with Observations Labeled by Grade Level. Students at a particular grade level have about the same age and show a lack of association between height and test score.

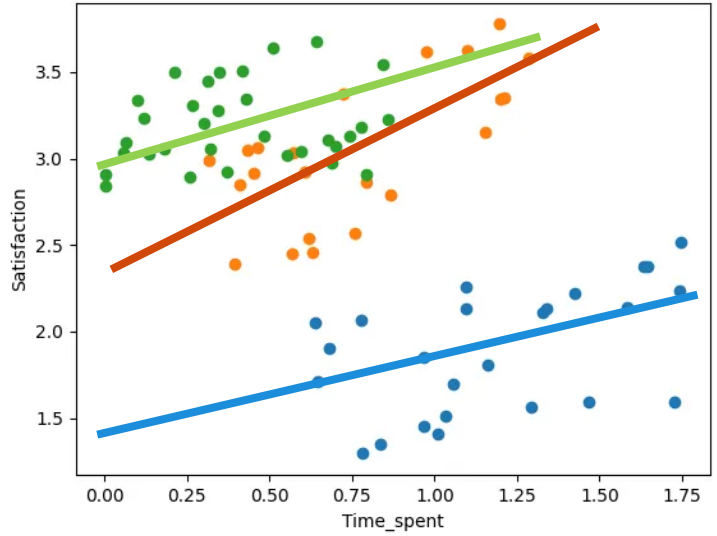


Simpson's paradox

- Describes the phenomenon when the bivariate association is **positive**, yet when controlling for a third variable, the association is **negative** at all values (or vice versa)



- *Example: is spending more time on a certain app associated with people's satisfaction with the app?*
- *Negative association in whole sample*
- *When controlling for country, positive relationship in each group!*



Multiple regression

- Response variable Y, explanatory variables $X_1, X_2, \ldots, X_k$ (all quantitative)

- Multiple regression equation:

$$\mu_y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

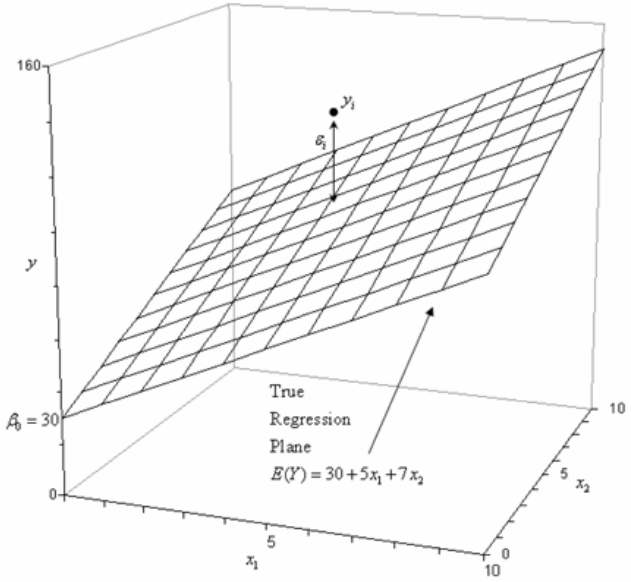
- For each explanatory variable, there is a **partial regression coefficient** β

Parameter interpretation

- Similar to last week univariate regression
- α is the intercept, and is the expected value of Y when all $X_i = 0$
- β_i are the partial regression coefficients and describe the change in $E[Y]$ when X_i increases by 1 unit, **and all other X are held constant**

- Rather than a regression line, we get a regression plane

  (or even hyperplane)



Prediction equation

- Still obtained by least squares, minimizing residual sum of squares
- $Residual\ SS = \sum(y_i - \hat{y}_i)^2$ (same as last week)

- Prediction for a single observation: $\hat{y} = a + b_1x_1 + b_2x_2 + \cdots + b_kx_k$

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Example: mental impairment study

- Y: mental impairment score (higher = worse mental health)
- X_1: life events score
- X_2: socioeconomic status

- Prediction equation using only X_1: $\hat{y} = 23.3 + 0.090x_1$
- Prediction equation using only X_2: $\hat{y} = 32.2 - 0.086x_2$

- Using multiple regression: $\hat{y} = 28.2 + 0.103x_1 - 0.097x_2$

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Example

- $\hat{y} = 28.2 + 0.103x_1 - 0.097x_2$

- **Interpretation**:
- For someone with 0 life events and SES of 0, predicted mental impairment is equal to the intercept a, which is 28.2
- For each 1-unit increase in life events score, *keeping SES constant,* predicted mental impairment increases by 0.103
- For each 1-unit increase in SES, *keeping life events constant*, predicted mental impairment decreases by 0.097

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Comments

- Partial effects in multiple regression **control** for other variables in the model, whereas effects in bivariate models **ignore** all other variables

- If the correlation between X_1 and X_2 is 0, the partial effects are the same as the bivariate effects (in a sample, correlation is almost never exactly 0)

Controlling for other variables

- $\hat{y} = 28.2 + 0.103x_1 - 0.097x_2$
- For $x_1 = 0$:
  - $\hat{y} = 28.2 - 0.097x_2$
- For $x_1 = 10$:
  - $\hat{y} = (28.2 + 0.103 * 10) - 0.097x_2 = 29.23 - 0.097x_2$

- This assumes that the partial slope for a variable **does not change** when values for other explanatory variables change! -> interaction later

Multiple correlation and R^2

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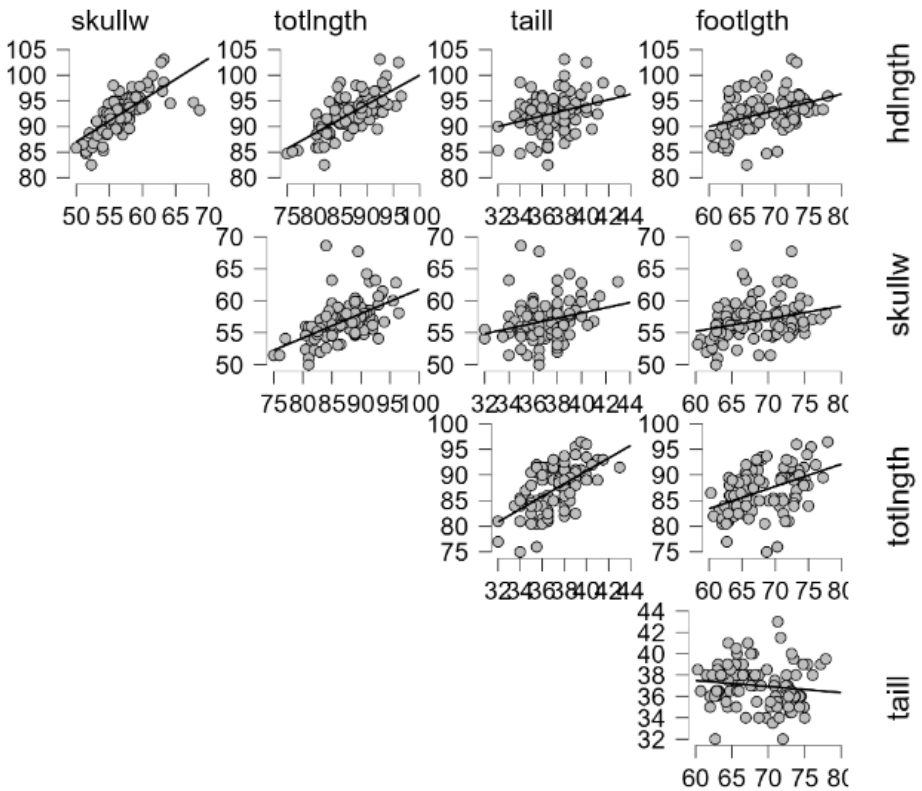
Correlation matrix

- Shows pairwise correlation between all variables
- Also check marginal distributions and scatterplots!

	hdlngth	skullw	totlngth	taill	footlgth
hdlngth		0.711	0.691	0.287	0.392
skullw	0.711		0.526	0.256	0.275
totlngth	0.691	0.526		0.566	0.445
taill	0.287	0.256	0.566		-0.126
footlgth	0.392	0.275	0.445	-0.126	



Multiple correlation (R)

- In bivariate linear regression, correlation r is the correlation between X and Y
- In multivariate linear regression, the **multiple correlation** R is the correlation between (the combination of all X variables) and (Y)
- In other words, R is the correlation between observed Y and predicted Y values
  - Correlation between Y and a single X variable in multiple regression is partial correlation

- Multiple correlation says how well the explanatory variables in the model are able to predict Y

R^2

- The **coefficient of multiple determination**, R^2, is the proportional reduction in error when using the prediction equation to predict Y, compared to using $\bar{y}$ to predict Y

- Same as in bivariate regression!
- $R^2 = \frac{Total\ SS - Residual\ SS}{Total\ SS} = \frac{\sum(y-\bar{y})^2 - \sum(y-\hat{y})^2}{\sum(y-\bar{y})^2}$

Example: mental impairment

- Predicting possum tail length from foot length and head length
- Software outputs table below
- $R^2 = \frac{Total\ SS - Residual\ SS}{Total\ SS} = \frac{395.228 - 336.248}{395.228} = 0.149$
- Foot length and head length explain 14.9% of variation in tail length
- Multiple correlation $R = \sqrt{0.149} = 0.386$ (sign always positive!)

ANOVA ▼

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	58.980	2	29.490	8.770	< .001
	Residual	336.248	100	3.362		
	Total	395.228	102			

Note. M_1 includes footlgth, hdlngth
Note. The intercept model is omitted, as no meaningful information can be shown.

Model Summary - taill ▼

Model	R	R^2	Adjusted R^2	RMSE
M_0	0.000	0.000	0.000	1.968
M_1	0.386	0.149	0.132	1.834

Note. M_1 includes footlgth, hdlngth

Properties of R and R^2

- $0 \leq R^2 \leq 1$ and $0 \leq R \leq 1$ (note difference with r!)
- Larger values means better predictions
- $R^2 = 1$ occurs only when all $y_i = \hat{y}_i$
- $R^2 = 0$ occurs only when all $y_i = \bar{y}$, then $Total\ SS = Residual\ SS$
  - Happens only when all $b_i = 0$ and $r = 0$ between Y and each X
- R^2 can never decrease when predictors are added
- With a single explanatory variable, $R^2 = r^2$ and $R = |r|$

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Inference: multiple regression

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Inference for multiple regression

- Significance test for a single β: last week
- Significance test for collective influence of explanatory variables: F-test

1. Assumptions

- Data were gathered using randomization
- There is a linear relationship between the set of Xs and Y in the population
- The conditional distribution of Y is normal AND has the same σ at each **combination of Xs**

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Inference for multiple regression

2. Hypotheses

Instead of testing a single β, we test them all

$H_0: \beta_i = 0$ for all i

$H_a: \beta_i \neq 0$ for at least one i

One-sided testing not possible

Inference for multiple regression

3. Test statistic

$$F = \frac{R^2/k}{(1-R^2)/(n-(k+1))} = \frac{Mean\ square\ for\ regression}{Mean\ square\ error}$$

where k is the number of explanatory variables

- When H_0 is true, F-values follow the F-distribution
- Larger R results in larger F (with constant n)
- P-value is the right-tail probability under F

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	58.980	2	29.490	8.770	< .001
	Residual	336.248	100	3.362		
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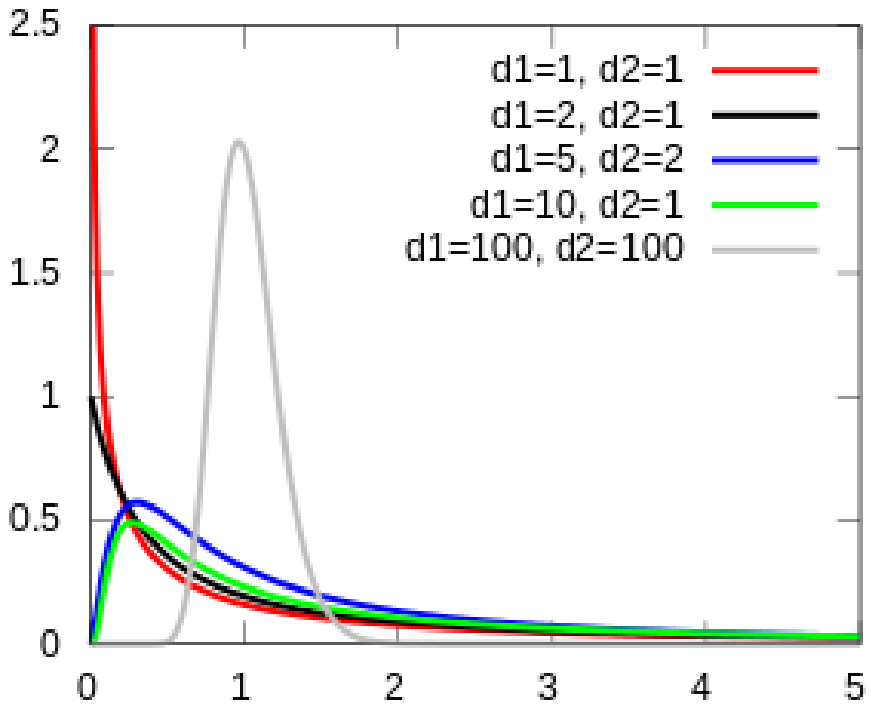
Note. M_1 includes footlgth, hdlngth

F-distribution



- Only non-negative values
- Distribution is skewed right
- Shape depends on two *df* values!!
  - $df_1 = k$ (# explanatory variables)
  - $df_2 = n - (k + 1)$ (sample size minus # estimated parameters)
- Mean is approximately 1 (closer to 1 as n increases)

$$F = \frac{R^2/k}{(1 - R^2)/(n - (k + 1))}$$

- Intuition: numerator is "explained error per parameter", denominator is "remaining error per observation", taking both *df* into account

Inference for multiple regression

4. P-value

F-values follow F-distribution, we need the F-table

Rows: $df_1 = k$

Columns: $df_2 = n - (k + 1)$

Cells: critical F-value for $\alpha = 0.05$

(other table for other α!)

$p < 0.05$ if $F > F_{crit}$

5. Conclusion: association?

***F* Distribution: Critical Values of *F* (5% significance level)**

ν_2 \ ν_1	1	2	3	4	5	6	7	8	9	10	12	14	16	18	20
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88	240.54	241.88	243.91	245.36	246.46	247.32	248.01
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.42	19.43	19.44	19.45
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.71	8.69	8.67	8.66
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.87	5.84	5.82	5.80
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.64	4.60	4.58	4.56
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.96	3.92	3.90	3.87
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.53	3.49	3.47	3.44
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.24	3.20	3.17	3.15
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.03	2.99	2.96	2.94
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.86	2.83	2.80	2.77
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.74	2.70	2.67	2.65
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.64	2.60	2.57	2.54
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.55	2.51	2.48	2.46
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.48	2.44	2.41	2.39
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.42	2.38	2.35	2.33
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.37	2.33	2.30	2.28
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.33	2.29	2.26	2.23
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.29	2.25	2.22	2.19
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.26	2.21	2.18	2.16
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.22	2.18	2.15	2.12
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.20	2.16	2.12	2.10
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.17	2.13	2.10	2.07
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.15	2.11	2.08	2.05
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.13	2.09	2.05	2.03
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.11	2.07	2.04	2.01
26	4.22	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27	2.22	2.15	2.09	2.05	2.02	1.99
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25	2.20	2.13	2.08	2.04	2.00	1.97
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24	2.19	2.12	2.06	2.02	1.99	1.96
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22	2.18	2.10	2.05	2.01	1.97	1.94
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.04	1.99	1.96	1.93

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Example: mental impairment

- We want to test if there is an association between the number of life events (x_1), SES (x_2) and mental impairment score (y)

1. **Assumptions**
2. **Hypotheses**

At $\alpha = 0.05$, we want to test:

$H_0: \beta_1 = \beta_2 = 0$

$H_a: \beta_1 \neq 0$ and/or $\beta_2 \neq 0$

Example: mental impairment

3. Test statistic

Calculate the F-statistic: $F = \frac{R^2/k}{(1-R^2)/(n-(k+1))}$

$$R^2 = \frac{Total\ SS - Residual\ SS}{Total\ SS} = \frac{1162.4-768.2}{1162.4} = 0.34$$

(34% of mental impairment explained)

$$F = \frac{0.34/2}{(1-0.34)/(20-(2+1))} = 9.495$$

	Sum of squares	df	MSE	F	P
Residual	768.2	2	197.1	9.495	<0.001
Regression	394.2	37	20.8		
Total	1162.4	39			

Example: mental impairment

4. P-value

For $(2, 37)$ degrees of freedom,
the critical value of F is 3.27

We found $F = 9.495$, so $p < 0.05$

5. Conclusion

There is a statistically significant association between number of life events, SES and mental impairment score

F-distribution: critical values of F at $\alpha = 0.05$

ν_2 \ ν_1	1	2	3	4	5	6	7	8
1	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07
35	4.12	3.27	2.87	2.64	2.49	2.37	2.29	2.22
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18
50	4.03	3.18	2.79	2.56	2.40	2.29	2.20	2.13

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Inference on individual β

- The multiple inference test only tells us that at least one of the $\beta \neq 0$
- But which? Or both?
- Follow-up test for partial effect of X_1, keeping X_2 fixed:
- $H_0: \beta_1 = 0$ versus $H_a: \beta_1 \neq 0$
- Test statistic same as last week: $t = \frac{b_i - 0}{se}$, formula for se is complex
- $df = n - (k + 1)$ (instead of $n - 2$ last week, when k was 1)

- Or use a CI for β_i, with the form $b_i \pm t * se$

Residual plots

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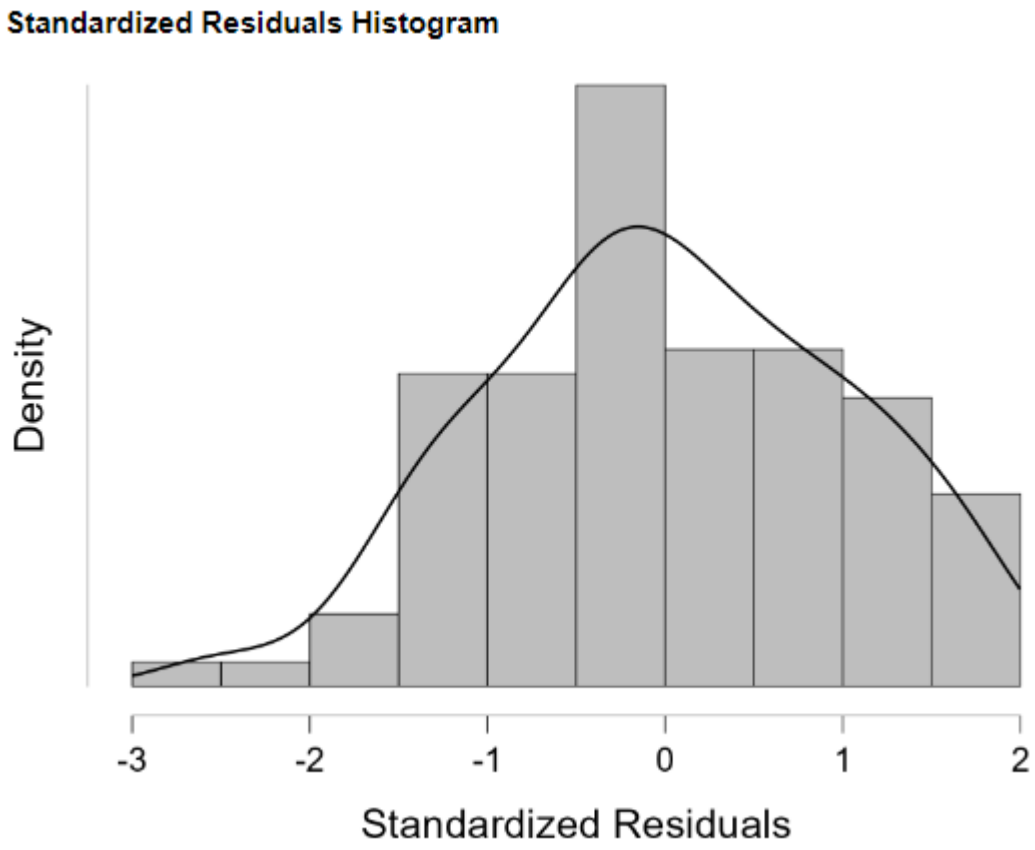
Residual plots

- Histogram of standardized residuals
- Residuals against response variable
- Residuals against each explanatory variable
- Residuals against predicted values
- Q-Q plot

Histogram

- Standardized residuals
- Should roughly all be between -3 and 3
- Normal distribution, symmetry

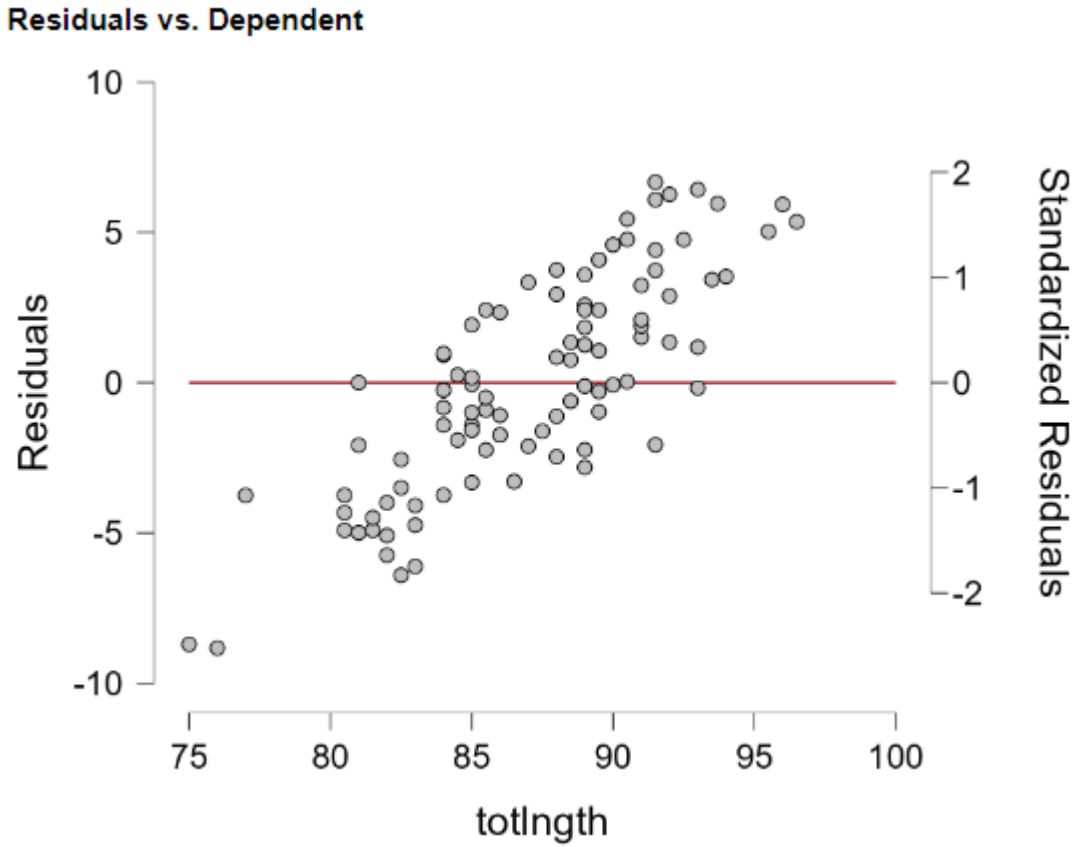
- Here: skewed left, meaning more positive residuals than negative residuals, the model is underestimating more than overestimating
- Positive residual = model predicts too low
- Negative residual = model predicts too high



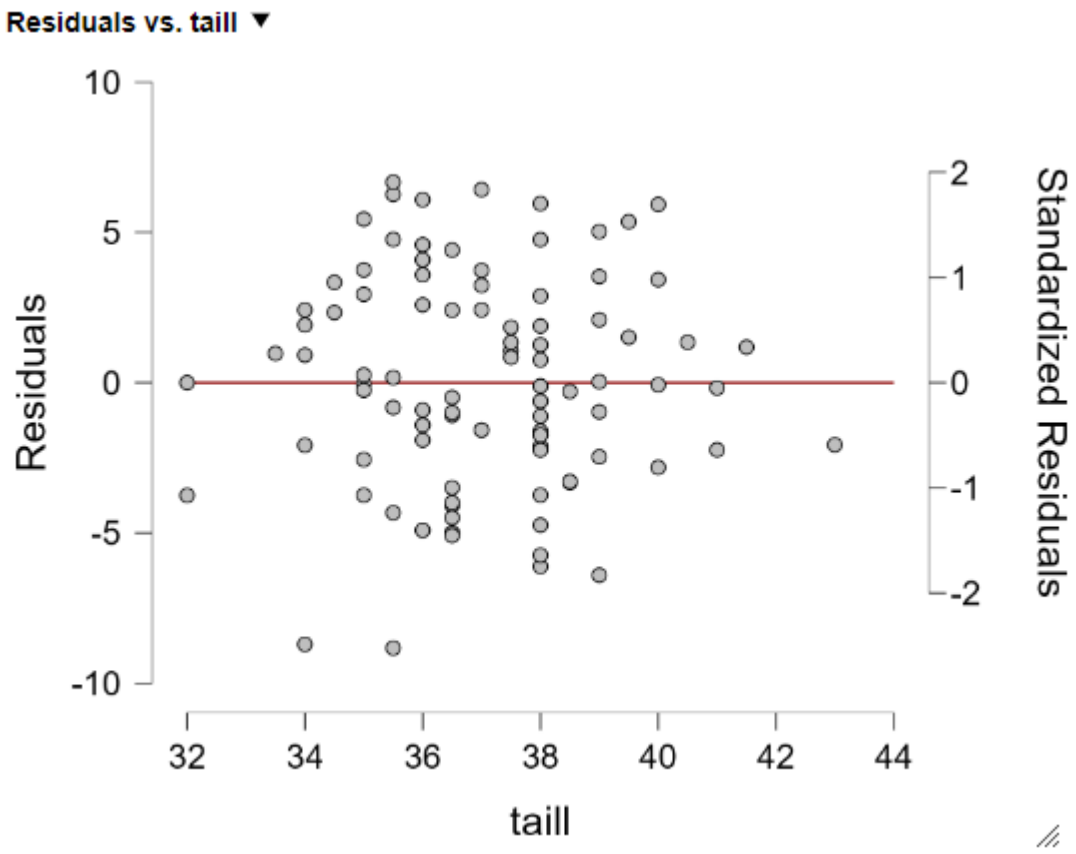
Residuals vs response

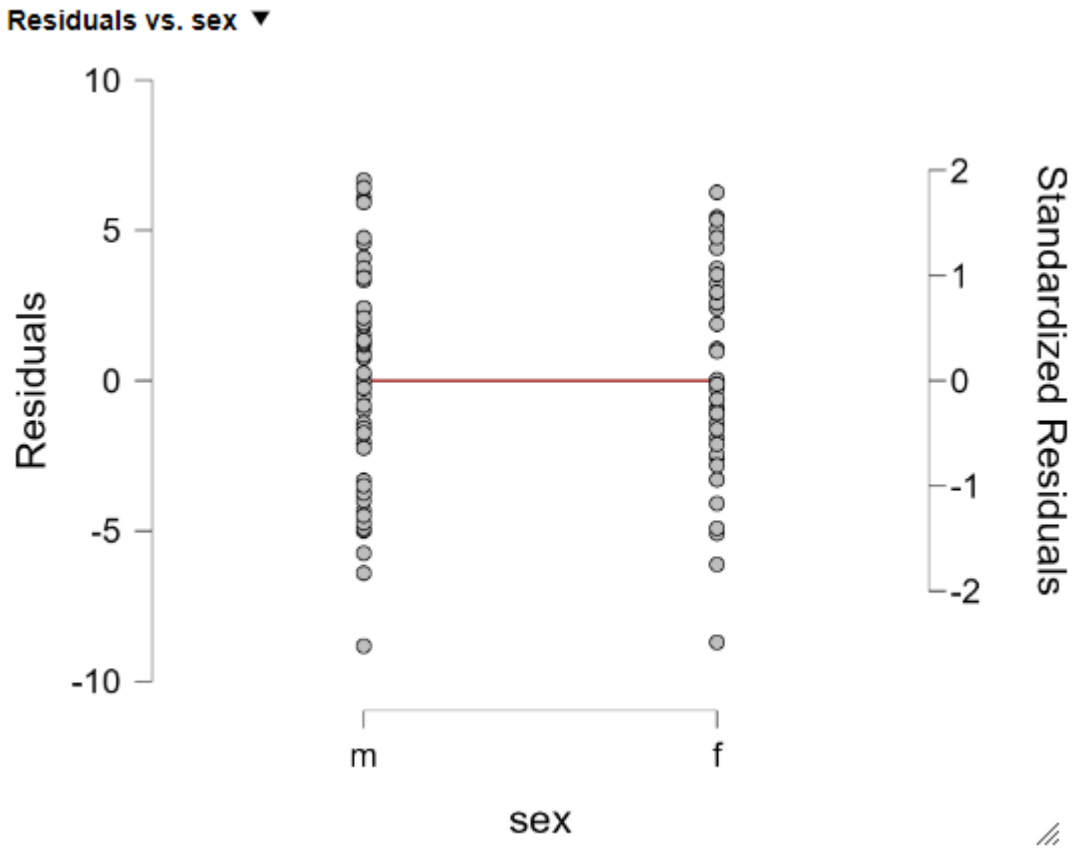
- Should be randomly scattered around $x = 0$ with same standard deviation everywhere

- Here: positive correlation between totlength and residuals. The model predicts too high for small possums and too low for large possums (meaning this model doesn't capture all the variation!)



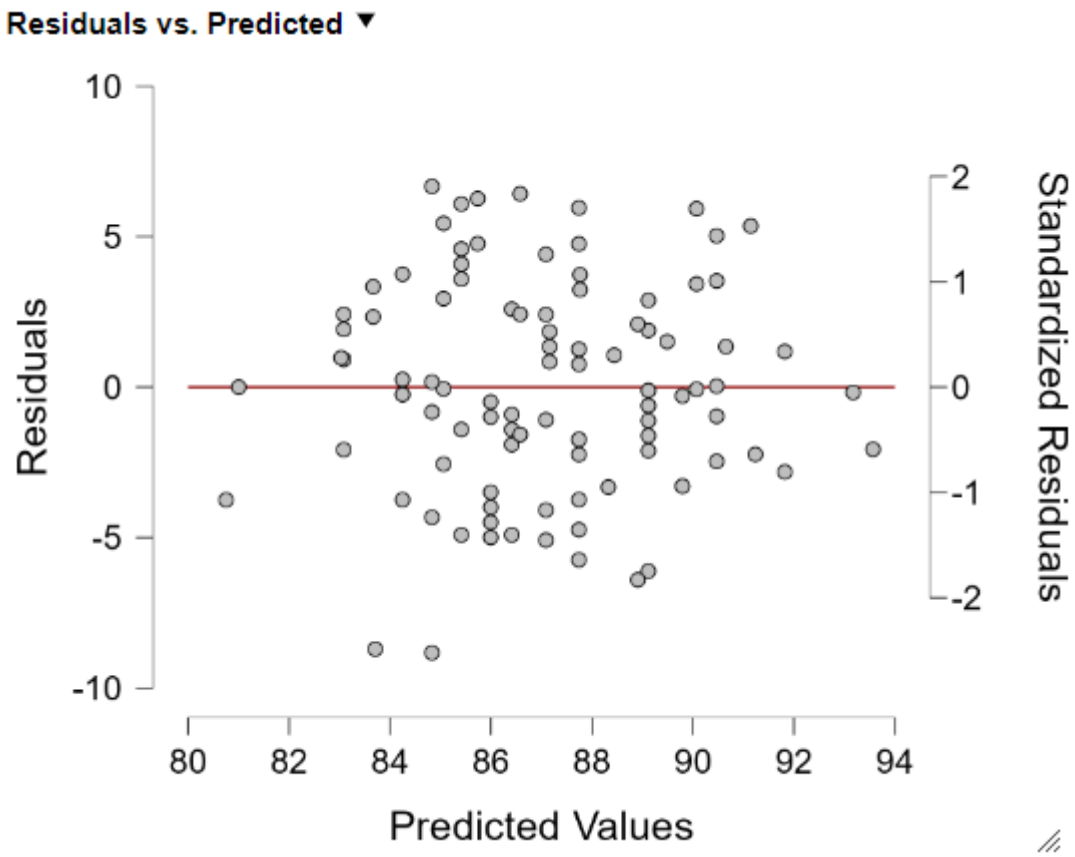
Residuals vs explanatory





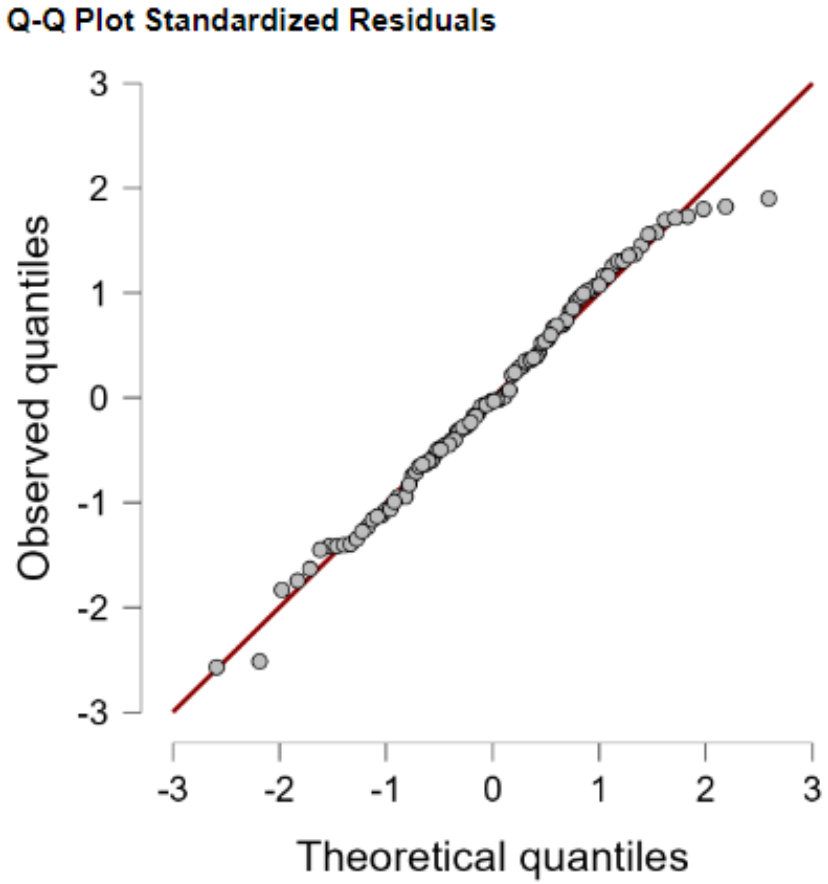
Residuals vs predicted

- Should be randomly scattered around $x = 0$

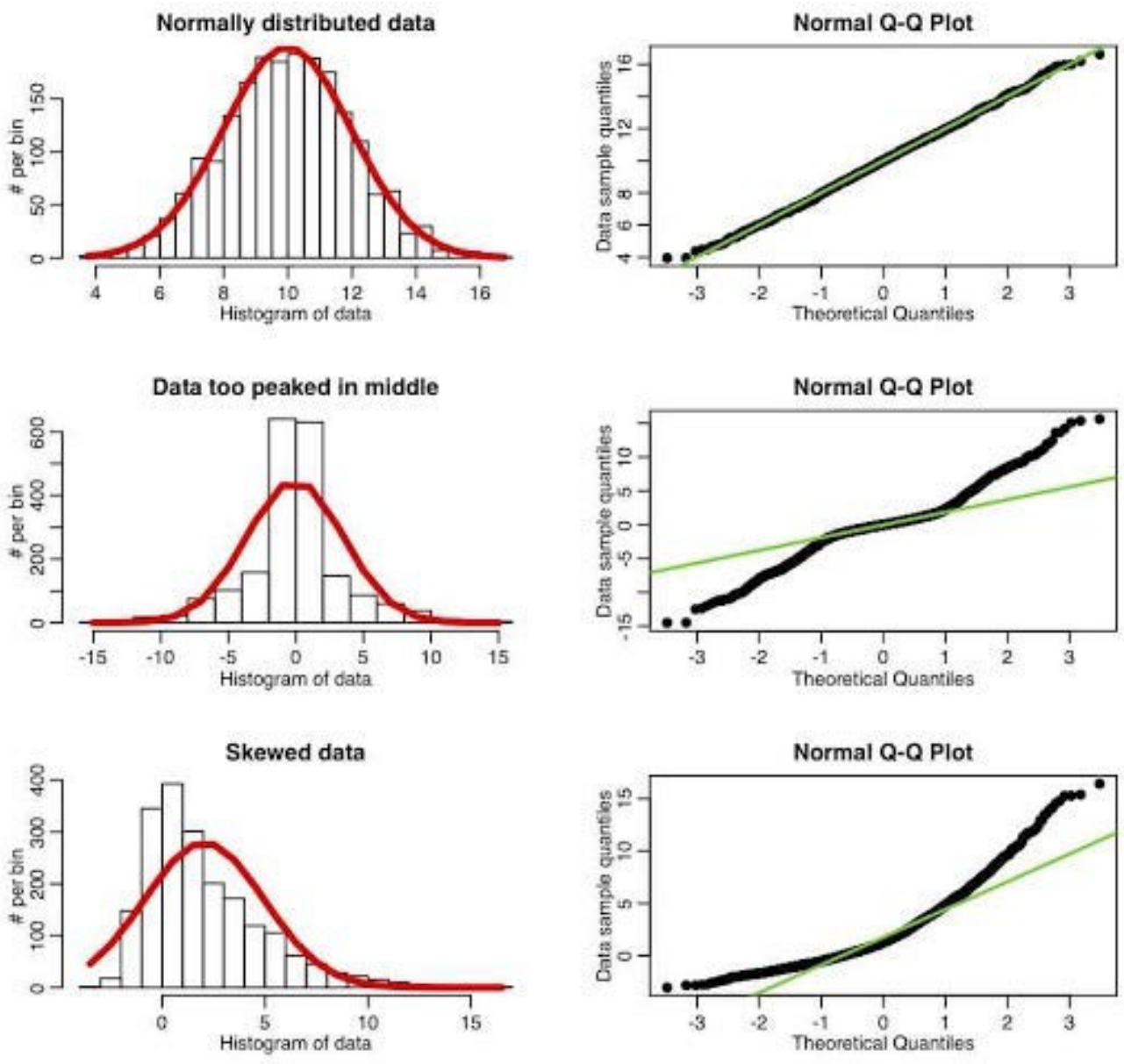


Q-Q plot

- Related to histogram of residuals
- Comparison to normal distribution
- Watch for deviations from the line at the extreme values!



Q-Q plot


Normally distributed data
per bin
Histogram of data
Normal Q-Q Plot
Data sample quantiles
Theoretical Quantiles
Data too peaked in middle
per bin
Histogram of data
Normal Q-Q Plot
Data sample quantiles
Theoretical Quantiles
Skewed data
per bin
Histogram of data
Normal Q-Q Plot
Data sample quantiles
Theoretical Quantiles

Interaction and categorical predictors

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Indicator variables

- Predicting possum total length from only sex (male/female):
  - $\hat{y} = 86.511 + 1.396 * sex_F$
  - sex_F is indicator variable, either 0 or 1. 1.396 gets added to the prediction for female possums

- Categorical predictor with N possible values needs N-1 indicator variables
  - $\hat{y} = 10 + 4 * color_R + 3 * color_G$
  - Predicted values for colors R, G and B?

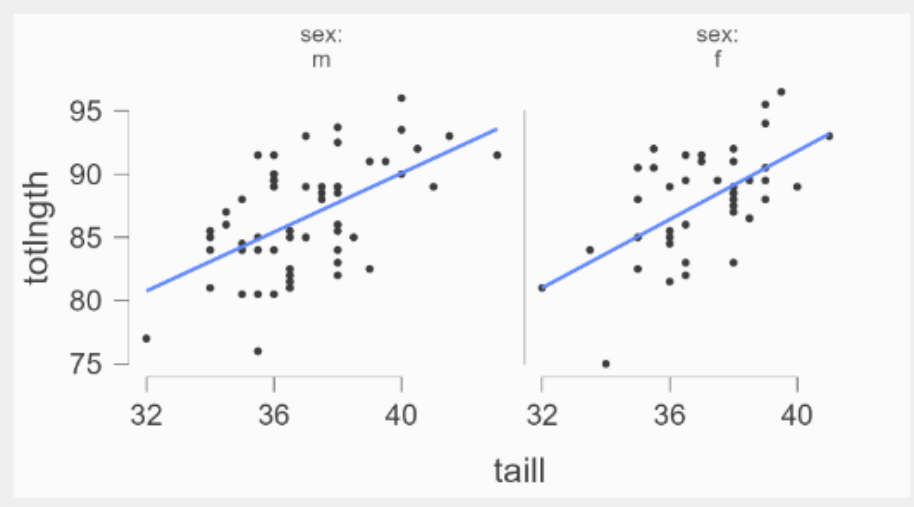
Coefficients ▼

Model		Unstandardized	Standard Error	Standardized[a]	t	p
M_0	(Intercept)	87.088	0.423		206.037	< .001
M_1	(Intercept)	86.511	0.547		158.028	< .001
	sex (f)	1.396	0.851		1.639	0.104

[a] Standardized coefficients can only be computed for continuous predictors.

Indicator and interaction

- Predicting possum length from tail length and sex:
  - $\hat{y} = 41 + 1.196 * sex_F + 1.232 * taill$
  - Partial slope for tail same for both sexes! But are they?



- Adding an interaction effect:
  - $\hat{y} = 43.47 - 5.751 * sex_F + 1.165 * taill + 0.188 * sex_F * taill$

Coefficients ▼

Model		Unstandardized	Standard Error	Standardized[a]	t	p
M_0	(Intercept)	87.088	0.423		206.037	< .001
M_1	(Intercept)	41.000	6.597		6.215	< .001
	sex (f)	1.196	0.705		1.695	0.093
	taill	1.232	0.178	0.560	6.916	< .001
M_2	(Intercept)	43.470	8.247		5.271	< .001
	taill	1.165	0.223	0.530	5.227	< .001
	sex (f)	−5.751	13.848		−0.415	0.679
	taill ✻ sex (f)	0.188	0.373		0.502	0.617

[a] Standardized coefficients can only be computed for continuous predictors.

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Multiple interaction terms

- With two predictors, only one interaction term is possible
- With more predictors, the number of possible interaction terms grows quickly:

Predictor variables	2-way interactions	3-way interactions	4-way interactions
X_1, X_2	$\beta_3 X_1 X_2$		
X_1, X_2, X_3	$\beta_4 X_1 X_2$ $\beta_5 X_1 X_3$ $\beta_6 X_2 X_3$	$\beta_7 X_1 X_2 X_3$	
X_1, X_2, X_3, X_4	$\beta_5 X_1 X_2$ $\beta_6 X_1 X_3$ $\beta_7 X_1 X_4$ $\beta_8 X_2 X_3$ $\beta_9 X_2 X_4$ $\beta_{10} X_3 X_4$	$\beta_{11} X_1 X_2 X_3$ $\beta_{12} X_1 X_2 X_4$ $\beta_{13} X_1 X_3 X_4$ $\beta_{14} X_2 X_3 X_4$	$\beta_{15} X_1 X_2 X_3 X_4$

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Logistic regression

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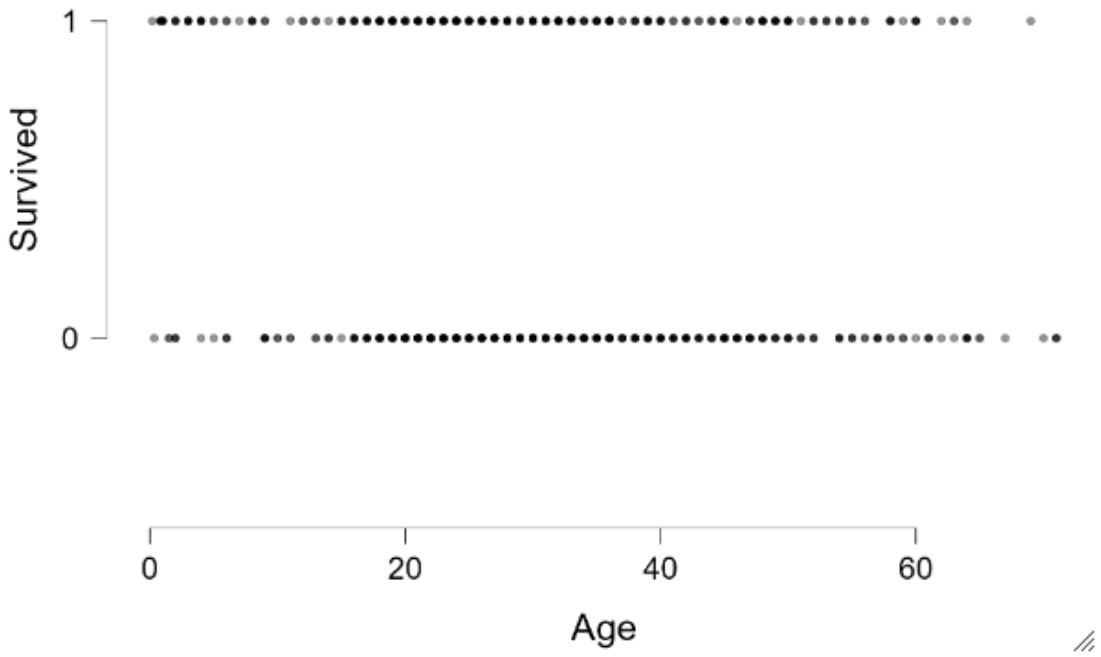
Logistic regression

- If Y is categorical (binary) we can use regression to model the probability of an outcome value
- Instead of a straight line, we need all probabilities between 0 and 1!
- $p = \frac{e^{\alpha+\beta x}}{1+e^{\alpha+\beta x}}$
- $\beta > 0$ means positive association with p
- Not a slope!



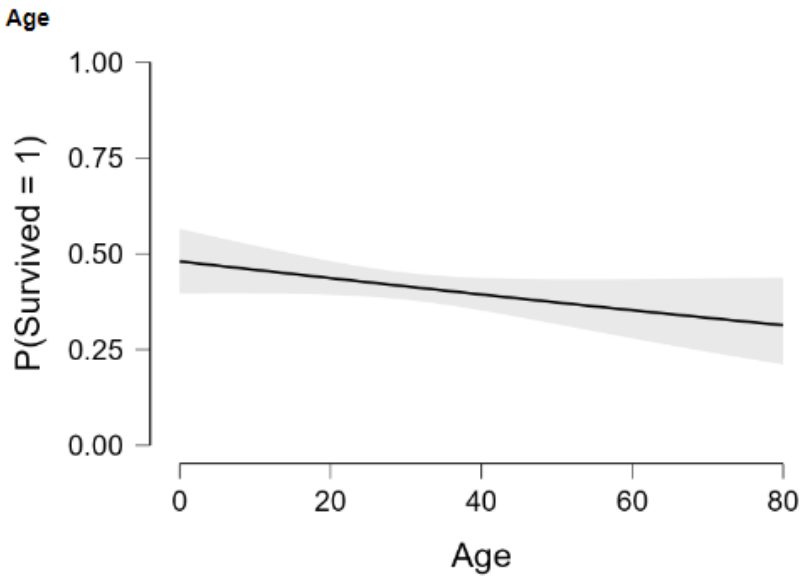
Interpreting logistic regression

- Compare estimate of p at two values (e.g., min and max of x)
  - If $\hat{p} = 0.5$ for age 0, and $\hat{p} = 0.32$ for age 80, is that a strong relationship between age and survival? (titanic dataset)
- At which value of x is $\hat{p} = 0.5$? Or, for which age are both outcomes equally likely?
  - For $x = -\alpha/\beta$
  - In this case, $\frac{0.081}{-0.009} = -9$!

Coefficients ▼

Model		Estimate	Standard Error	Odds Ratio	z	Wald Test		
						Wald Statistic	df	p
M_0	(Intercept)	−0.347	0.074	0.707	−4.704	22.131	1	< .001
M_1	(Intercept)	−0.081	0.174	0.922	−0.468	0.219	1	0.640
	Age	−0.009	0.005	0.991	−1.681	2.826	1	0.093

Note. Survived level '1' coded as class 1.



Wrapping up

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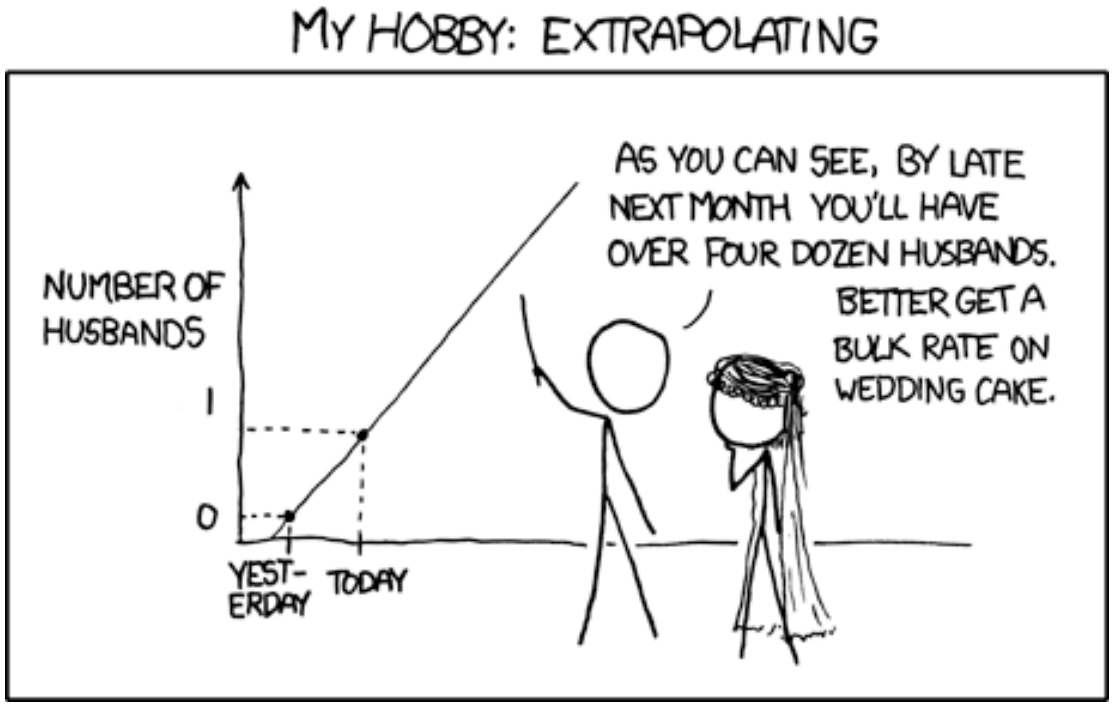
Wrapping up

- Read chapter 13
- JASP tutorial multiple regression

- **Assignment 2 is online**
- Deadline: 7 December 23:59h

- Next week: nonparametric statistics



https://xkcd.com/605/